

FIREWALL & NETWORK ACCESS CONTROL REVIEW REPORT

ICT Department — Friday Morning Cybersecurity Practical Assignment

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Date:

12th June 2026

1. Introduction

This practical assignment focused on reviewing firewall configurations and network access control settings within the organization's network infrastructure. The objective was to determine whether unauthorized access is properly restricted and whether existing firewall rules adequately protect the network environment.

The assessment involved:

- Network scanning
- Router and firewall review
- Wi-Fi access control analysis
- Open port identification
- Firewall security recommendations

2. Objectives

The purpose of this task was to:

- Understand how firewalls protect computer networks
- Evaluate current network access control mechanisms
- Identify exposed services and potential risks
- Recommend improvements to strengthen network security

3. Practical Activities Performed

3.1 Network Discovery & Access Review

Command Used

```
nmap -sn 192.168.0.0/24
```

Devices Identified

The following active devices were discovered on the network:

IP Address	Device Type
192.168.0.1	Tenda Router
192.168.0.100	TP-Link Device
192.168.0.110	Dell Workstation
192.168.0.111	Intel-Based Device
192.168.0.102–107	Unknown Devices
192.168.0.112–114	Unknown Devices

4. Firewall & Port Exposure Review

A. Router Assessment

Command Used

```
nmap -sV 192.168.0.1
```

Findings

Port	Status	Service
80/tcp	OPEN	GoAhead WebServer

Analysis

The router management interface was accessible over HTTP.

Security Risk

- HTTP is not encrypted
- Login credentials could potentially be intercepted on insecure networks
- Router admin panel exposure increases risk of unauthorized access

Recommendation

- Disable HTTP management access
- Enable HTTPS-only administration
- Restrict router management access to administrators only
- Change default router credentials
- Enable WPA3 wireless encryption

B. Workstation Assessment

Command Used

```
nmap -sV 192.168.0.110
```

Findings

Port	Status	Service
3389/tcp	OPEN	Microsoft Remote Desktop (RDP)

Analysis

Remote Desktop Protocol (RDP) was enabled on the workstation.

Security Risk

- RDP is a common target for brute-force attacks
- Weak passwords may lead to unauthorized remote access

Recommendation

- Restrict RDP access to authorized administrators only
- Enable strong passwords and MFA
- Use firewall rules to limit allowed IP addresses
- Disable RDP when not required

C. Filtered Devices

Commands Used

```
nmap -sV 192.168.0.100  
nmap -sV 192.168.0.111
```

Findings

All ports were filtered or blocked.

Analysis

This indicates the firewall is actively restricting unauthorized scans and external access.

Security Benefit

- Reduces attack surface
- Prevents unauthorized service enumeration
- Improves network protection

5. Wi-Fi Access Control Review

Current Observations

The wireless network was protected and required authentication before access.

Existing Controls Observed

- Password-protected wireless access

- Internal private IP addressing
- Router-based device management

Identified Weaknesses

Weakness	Risk
Unknown connected devices	Unauthorized access
Router HTTP management enabled	Credential interception
No visible network segmentation	Lateral movement risk

6. Recommended Network Access Control Improvements

Recommended Enhancements

A. Wireless Security

- Upgrade Wi-Fi encryption to WPA3
- Use strong Wi-Fi passwords
- Disable WPS functionality
- Regularly change wireless passwords

B. Device Access Control

- Implement MAC address filtering
- Remove unknown devices from network
- Use separate guest Wi-Fi network
- Segment staff and administrative systems

C. Firewall Improvements

- Close unused ports
- Restrict remote administration
- Enable firewall logging
- Configure intrusion detection alerts
- Block unnecessary inbound traffic

7. External Port Exposure Review

Firewall Testing Concept

Firewall exposure testing was researched using:

- [GRC ShieldsUP!](https://www.grc.com/shieldsup/)

Purpose

The tool helps determine:

- Whether ports are visible externally

- If firewall rules properly block incoming connections
- Whether the network appears “stealth” to attackers

Recommended Future Action

Conduct a full external firewall scan from the public internet connection.

8. Lessons Learned

During this practical assignment, the following skills and concepts were learned:

- Basic firewall evaluation techniques
- Identification of open and filtered ports
- Understanding network access control
- Importance of securing router administration interfaces
- Risks associated with exposed RDP services
- Importance of wireless security best practices

9. Conclusion

The firewall and network access review revealed that some protection mechanisms are already active, particularly through filtered ports on several devices. However, improvements are still needed to strengthen the network against unauthorized access.

The major risks identified include:

- Router management exposure over HTTP
- Open RDP service
- Lack of visible network segmentation
- Presence of unknown devices on the network

Implementing the recommended firewall hardening measures, network segmentation, stronger wireless security, and access restrictions will significantly improve the organization’s overall cybersecurity posture.